Personal Statement — MSc in Artificial Intelligence, University of Macau

On the multi-robot platform I develop for at Kaihong, a high-level task plan cannot be sent directly to a robot. It must pass through an action-proposal, policy-check, execution and state-feedback chain before any actuator is allowed to move. Building that chain has been the most instructive part of my robotics work so far, because it is the point where an AI decision becomes a physical action, and designing it made the gap in my training clear. I can make a robot balance, navigate and follow a trajectory, and I can deploy a vision model to an edge device, but my experience with machine learning has so far been mainly practical; I now need to understand its foundations, failure modes and system-level safeguards. The MSc in Artificial Intelligence is where I intend to close that gap.

I am a final-year Computer Science student at Shenzhen Technology University, ranked 7 of 156, with control theory courses added to the core curriculum. My AI experience began with a personal project, an OpenHarmony smart gardening system, in which I built a YOLOv5 inference chain quantised to INT8 on RK3588, a LangGraph decision chain with a rule-based fast path for extreme states, and a device-side irrigation state machine, and integrated them into a closed loop from camera to pump. Despite its small scale, it exposed questions that become more important in larger systems: how to interpret a detector's confidence, when a learned policy should yield to a rule, and how to keep a physical actuator safe when the decision module is uncertain.

The rest of my experience is in robot control and navigation on real hardware. At the Shenzhen Institute of Artificial Intelligence and Robotics for Society I was responsible for the balance control of a dual-arm ballbot, combining a deployed cascaded PID controller with a linearised model and LQR analysis, and the chassis maintained balance while the arms moved. In laboratory projects I implemented gain-scheduled LQR and virtual model control on a wheel-legged robot and the full localisation, planning and MPC tracking chain on a car-like vehicle. At Future Power I worked on VR teleoperation of a dual-arm robot with constrained inverse kinematics. Together, these projects showed me the limits of fixed pipelines: I tuned parameters by hand, fused states with fixed filters and used planners that assumed a largely static environment. Learning-based perception, state estimation and control are the next direction I want to pursue, but I want to do so with a proper understanding of their mathematical foundations and failure modes.

The MSc in Artificial Intelligence at the University of Macau would help me move from practical deployment to a more systematic understanding of AI methods. I have used computer vision, edge inference and an agent decision chain in a working robot-related application, and I have also seen how perception and decision quality affect control and actuation on real hardware. I now want stronger foundations in machine learning, data science, information processing and intelligent systems so that I can evaluate models, understand their limitations and integrate them responsibly into physical systems.

My goal is to work as a robotics algorithm engineer, combining perception, planning and control with learning-based methods. The programme would complement my existing hands-on experience with a broader AI foundation, while my background would allow me to test learned components against the timing, uncertainty and safety constraints of real robots.